ASSIGNMENT - 01

APACHE HIVE

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Scenario

A retail organization stores transaction records as CSV files in HDFS. The analytics team needs structured SQL-style access to study sales and revenue patterns across product categories.

Problem Statement

Create an external Hive table over transaction data stored in HDFS. Use HiveQL to handle missing values, compute total sales revenue, and determine the average order value for each product category.

Apache Hive & Hadoop

Apache Hive is a data warehouse and analysis component of the Hadoop ecosystem. It provides HiveQL, a SQL-like language for querying and analyzing information stored in HDFS.

Apache Hadoop is an open-source framework for distributed storage and processing of large datasets. HDFS provides storage, while MapReduce and YARN support processing and resource management.

Docker

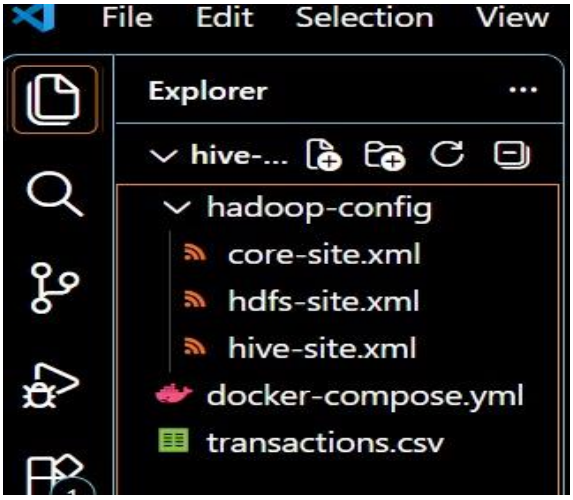
Docker provides an isolated and repeatable environment for running Hadoop and Hive. It is useful here because the required Hive setup uses Linux-oriented scripts and configuration. Running the services in containers reduces host-system configuration issues.

Workflow

Docker Desktop → Hadoop/HDFS → Transaction CSV → Apache Hive → External Table → Data Cleaning → HiveQL Aggregation → Revenue & Average Order Value.

Docker-Compose.yml

The Docker Compose setup launches the Hadoop and Hive components in separate containers. The NameNode manages HDFS metadata, the DataNode stores the actual blocks, and HiveServer2 provides the interface for running HiveQL commands. The transaction CSV is placed in HDFS, mapped through an external table, and then analyzed with Hive queries.



docker-compose.yml

```yaml
services:

  namenode:
    image: apache/hadoop:3.4.1
    container_name: hive-namenode
    hostname: namenode
    command: >
      bash -c "
      if [ ! -d /data/name/current ]; then
        hdfs namenode -format -force;
      fi;
      hdfs namenode
      "
    environment:
      HADOOP_HOME: /opt/hadoop
      HADOOP_CONF_DIR: /opt/hadoop/etc/hadoop
    volumes:
      - namenode_data:/data
      - ./hadoop-config/core-site.xml:/opt/hadoop/etc/hadoop/core-site.xml
      - ./hadoop-config/hdfs-site.xml:/opt/hadoop/etc/hadoop/hdfs-site.xml
    ports:
      - "9870:9870"
      - "9000:9000"


  datanode:
    image: apache/hadoop:3.4.1
    container_name: hive-datanode
    hostname: datanode
    depends_on:
      - namenode
    command: ["hdfs", "datanode"]
    environment:
      HADOOP_HOME: /opt/hadoop
      HADOOP_CONF_DIR: /opt/hadoop/etc/hadoop
    volumes:
      - datanode_data:/data
      - ./hadoop-config/core-site.xml:/opt/hadoop/etc/hadoop/core-site.xml
      - ./hadoop-config/hdfs-site.xml:/opt/hadoop/etc/hadoop/hdfs-site.xml

  hive:
    image: apache/hive:4.0.1
    container_name: hive-server
    hostname: hive-server
    depends_on:
      - namenode
      - datanode
    environment:
      SERVICE_NAME: hiveserver2
    ports:
      - "10000:10000"
      - "10002:10002"
volumes:
  namenode_data:
  datanode_data:
```

Transaction.csv

transactions.csv

```
transaction_id,product_name,category,quantity,price
T001,Laptop,Electronics,2,55000
T002,Mouse,Electronics,5,800
T003,Office Chair,Furniture,2,7500
T004,Notebook,Stationery,10,120
T005,Keyboard,Electronics,3,1500
T006,Desk,Furniture,1,12000
T007,Pen,Stationery,20,50
T008,Monitor,Electronics,2,18000
T009,Bookshelf,Furniture,1,9000
T010,Printer,,1,15000
T011,Headphones,Electronics,,2500
T012,Paper,Stationery,5,300
T013,Table,Furniture,2,6500
T014,Webcam,Electronics,2,2200
T015,Marker,Stationery,10,80
```

1. Checking Docker Status

```
D:\hive-assignment>docker compose ps
NAME            IMAGE                COMMAND                  SERVICE    CREATED       STATUS         PORTS
hive-datanode   apache/hadoop:3.4.1   "/usr/local/bin/dumb…"   datanode   9 hours ago   Up 5 seconds
hive-namenode   apache/hadoop:3.4.1   "/usr/local/bin/dumb…"   namenode   9 hours ago   Up 5 seconds   0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 0.0.0.0:
9870->9870/tcp, [::]:9870->9870/tcp
hive-server     apache/hive:4.0.1     "sh -c /entrypoint.sh"   hive       9 hours ago   Up 4 seconds   0.0.0.0:10000->10000/tcp, [::]:10000->10000/tcp, 0.0.
0.0:10002->10002/tcp, [::]:10002->10002/tcp
```

2. Connecting through Command Line Interface (CLI)

```
D:\hive-assignment>docker exec -it hive-server beeline -u jdbc:hive2://localhost:10000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://localhost:10000
Connected to: Apache Hive (version 4.0.1)
Driver: Hive JDBC (version 4.0.1)
Transaction isolation: TRANSACTION_REPEATABLE_READ
Beeline version 4.0.1 by Apache Hive
0: jdbc:hive2://localhost:10000>
```

3. Creating the Database

```
0: jdbc:hive2://localhost:10000> CREATE DATABASE retail_db;
INFO  : Compiling command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac): CREATE DATABASE retail_db
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac); Time taken: 0.186 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac): CREATE DATABASE retail_db
INFO  : Starting task [Stage-0:DDL] in serial mode
INFO  : Completed executing command(queryId=hive_20260925152520_0e262785-5ce6-49f6-a4ac-47a28d8cb1ac); Time taken: 0.052 seconds
No rows affected (0.309 seconds)
```

4. Creating the External Table

An external Hive table is created over the transaction data stored in HDFS. The table definition matches the CSV columns and uses comma delimiters.

```
0: jdbc:hive2://localhost:10000> CREATE EXTERNAL TABLE transactions (
. . . . . .>     transaction_id STRING,
. . . . . .>     product_name STRING,
. . . . . .>     category STRING,
. . . . . .>     quantity INT,
. . . . . .>     price DOUBLE
. . . . . .> )
. . . . . .> ROW FORMAT DELIMITED
. . . . . .> FIELDS TERMINATED BY ','
. . . . . .> STORED AS TEXTFILE
. . . . . .> LOCATION 'hdfs://namenode:9000/retail_data'
. . . . . .> TBLPROPERTIES ("skip.header.line.count"="1");
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM transactions;
INFO  : Compiling command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4): SELECT * FROM transactions
INFO  : No Stats for retail_db@transactions, Columns: transaction_id, quantity, price, category, product_name
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:transactions.transaction_id, type:string, comment:null), FieldSchema(name:tr
oduct_name, type:string, comment:null), FieldSchema(name:transactions.category, type:string, comment:null), FieldSchema(name:transactions.quant
, comment:null), FieldSchema(name:transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4); Time taken: 2.35 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4): SELECT * FROM transactions
INFO  : Completed executing command(queryId=hive_20260925152830_9a815af4-5926-473b-9710-aaa5d0edbfb4); Time taken: 0.001 seconds
+-----------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| transactions.transaction_id | transactions.product_name     | transactions.category       | transactions.quantity          | transactions.price          |
+-----------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| T001                        | Laptop                        | Electronics                 | 2                              | 55000.0                     |
| T002                        | Mouse                         | Electronics                 | 5                              | 800.0                       |
| T003                        | Office Chair                  | Furniture                   | 2                              | 7500.0                      |
| T004                        | Notebook                      | Stationery                  | 10                             | 120.0                       |
| T005                        | Keyboard                      | Electronics                 | 3                              | 1500.0                      |
| T006                        | Desk                          | Furniture                   | 1                              | 12000.0                     |
| T007                        | Pen                           | Stationery                  | 20                             | 50.0                        |
| T008                        | Monitor                       | Electronics                 | 2                              | 18000.0                     |
| T009                        | Bookshelf                     | Furniture                   | 1                              | 9000.0                      |
| T010                        | Printer                       |                             | 1                              | 15000.0                     |
| T011                        | Headphones                    | Electronics                 | NULL                           | 2500.0                      |
| T012                        | Paper                         | Stationery                  | 5                              | 300.0                       |
| T013                        | Table                         | Furniture                   | 2                              | 6500.0                      |
| T014                        | Webcam                        | Electronics                 | 2                              | 2200.0                      |
| T015                        | Marker                        | Stationery                  | 10                             | 80.0                        |
+-----------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
15 rows selected (2.811 seconds)
```

5. Identifying Missing Values and Cleaning the Table

The cleaning query checks for missing category or quantity values. A missing category is replaced with an appropriate default label, while a missing quantity is assigned a default value.

```
0: jdbc:hive2://localhost:10000> CREATE TABLE cleaned_transactions AS
. . . . . .> SELECT
. . . . . .>     transaction_id,
. . . . . .>     product_name,
. . . . . .>     CASE
. . . . . .>         WHEN category IS NULL OR TRIM(category) = '' THEN 'Unknown'
. . . . . .>         ELSE category
. . . . . .>     END AS category,
. . . . . .>     CASE
. . . . . .>         WHEN quantity IS NULL THEN 1
. . . . . .>         ELSE quantity
. . . . . .>     END AS quantity,
. . . . . .>     price
. . . . . .> FROM transactions;
```

```
0: jdbc:hive2://localhost:10000> SELECT *
. . . . . .> FROM transactions
. . . . . .> WHERE category IS NULL
. . . . . .>    OR TRIM(category) = ''
. . . . . .>    OR quantity IS NULL;
INFO  : Compiling command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821): SELECT *
FROM transactions
WHERE category IS NULL
OR TRIM(category) = ''
OR quantity IS NULL
INFO  : No Stats for retail_db@transactions, Columns: transaction_id, quantity, price, category, product_name
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:transactions.transaction_id, type:string, comment:null), FieldSchema
oduct_name, type:string, comment:null), FieldSchema(name:transactions.category, type:string, comment:null), FieldSchema(name:transactio
, comment:null), FieldSchema(name:transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821); Time taken: 0.505 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821): SELECT *
FROM transactions
WHERE category IS NULL
OR TRIM(category) = ''
OR quantity IS NULL
INFO  : Completed executing command(queryId=hive_20260925152851_4bbbaa32-aafc-4e0e-82a1-75026b4ef821); Time taken: 0.001 seconds
+-----------------------------+---------------------------+--------------------------+---------------------------+--------------------------+
| transactions.transaction_id | transactions.product_name | transactions.category    | transactions.quantity     | transactions.price       |
+-----------------------------+---------------------------+--------------------------+---------------------------+--------------------------+
| T010                        | Printer                   |                          | 1                         | 15000.0                  |
| T011                        | Headphones                | Electronics              | NULL                      | 2500.0                   |
+-----------------------------+---------------------------+--------------------------+---------------------------+--------------------------+
2 rows selected (0.562 seconds)
```

6. Calculating Total Sales Revenue

```
0: jdbc:hive2://localhost:10000> SELECT * FROM cleaned_transactions;
INFO  : Compiling command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a): SELECT * FROM cleaned_transactions
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:cleaned_transactions.transaction_id, type:string, comment:null), FieldSchema(name:cleaned_transactions.produc
:string, comment:null), FieldSchema(name:cleaned_transactions.category, type:string, comment:null), FieldSchema(name:cleaned_transactions.quantity, type:int, comment:null), Fie
e:cleaned_transactions.price, type:double, comment:null)], properties:null)
INFO  : Completed compiling command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a); Time taken: 0.1 seconds
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a): SELECT * FROM cleaned_transactions
INFO  : Completed executing command(queryId=hive_20260925153022_0033bd91-6f25-4a83-aff8-9025a86a763a); Time taken: 0.0 seconds
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| cleaned_transactions.transaction_id | cleaned_transactions.product_name | cleaned_transactions.category | cleaned_transactions.quantity | cleaned_transactions.price |
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
| T001                              | Laptop                        | Electronics                 | 2                              | 55000.0                     |
| T002                              | Mouse                         | Electronics                 | 5                              | 800.0                       |
| T003                              | Office Chair                  | Furniture                   | 2                              | 7500.0                      |
| T004                              | Notebook                      | Stationery                  | 10                             | 120.0                       |
| T005                              | Keyboard                      | Electronics                 | 3                              | 1500.0                      |
| T006                              | Desk                          | Furniture                   | 1                              | 12000.0                     |
| T007                              | Pen                           | Stationery                  | 20                             | 50.0                        |
| T008                              | Monitor                       | Electronics                 | 2                              | 18000.0                     |
| T009                              | Bookshelf                     | Furniture                   | 1                              | 9000.0                      |
| T010                              | Printer                       | Unknown                     | 1                              | 15000.0                     |
| T011                              | Headphones                    | Electronics                 | 1                              | 2500.0                      |
| T012                              | Paper                         | Stationery                  | 5                              | 300.0                       |
| T013                              | Table                         | Furniture                   | 2                              | 6500.0                      |
| T014                              | Webcam                        | Electronics                 | 2                              | 2200.0                      |
| T015                              | Marker                        | Stationery                  | 10                             | 80.0                        |
+-----------------------------------+-------------------------------+-----------------------------+--------------------------------+-----------------------------+
15 rows selected (0.129 seconds)
```

The total revenue is calculated by multiplying quantity by price for each transaction and summing the resulting values. The query produces a total sales revenue of 229900.0.

HiveQL used: SELECT SUM(quantity * price) AS total_sales_revenue FROM cleaned_transactions;

```
0: jdbc:hive2://localhost:10000> SELECT
. . . . . .>     SUM(quantity * price) AS total_sales_revenue
. . . . . .> FROM cleaned_transactions;
INFO  : Compiling command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e
SUM(quantity * price) AS total_sales_revenue
FROM cleaned_transactions
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:total_sales_reve
INFO  : Completed compiling command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e
SUM(quantity * price) AS total_sales_revenue
FROM cleaned_transactions
INFO  : Query ID = hive_20260925153052_979e6c9c-9136-4f6b-9790-9813e5011c4b
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260925153052_979e6c9c-9136-4
INFO  : Session is already open
INFO  : Dag name: SELECT
SUM(quantity *......aned_transactions (Stage-1)
INFO  : HS2 Host: [hive-server], Query ID: [hive_20260925153052_979e6c9c-9136-4f6b-
INFO  : Status: Running (Executing on YARN cluster with App id application_17903501

INFO  : Completed executing command(queryId=hive_20260925153052_979e6c9c-9136-4f6b-
+----------------------+
| total_sales_revenue  |
+----------------------+
| 229900.0             |
+----------------------+
1 row selected (0.886 seconds)
```

7. Calculating Average Order Value by Category

```
0: jdbc:hive2://localhost:10000> SELECT
. . . . . .>     category,
. . . . . .>     AVG(quantity * price) AS average_order_value
. . . . . .> FROM cleaned_transactions
. . . . . .> GROUP BY category;
INFO  : Compiling command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b): SELECT
category,
AVG(quantity * price) AS average_order_value
FROM cleaned_transactions
GROUP BY category
INFO  : Semantic Analysis Completed (retrial = false)
INFO  : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:category, type:string, comment:null), FieldSch
INFO  : Completed compiling command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b); Time taken
INFO  : Concurrency mode is disabled, not creating a lock manager
INFO  : Executing command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b): SELECT
category,
AVG(quantity * price) AS average_order_value
FROM cleaned_transactions
GROUP BY category
INFO  : Query ID = hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b
INFO  : Total jobs = 1
INFO  : Launching Job 1 out of 1
INFO  : Starting task [Stage-1:MAPRED] in serial mode
INFO  : Subscribed to counters: [] for queryId: hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b
INFO  : Session is already open
INFO  : Dag name: SELECT
category,
AVG(......GROUP BY category (Stage-1)
INFO  : HS2 Host: [hive-server], Query ID: [hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b], Dag ID: [da
INFO  : Status: Running (Executing on YARN cluster with App id application_1790350157453_0001)

INFO  : Completed executing command(queryId=hive_20260925153123_bdac5e29-0157-4ad9-8cef-156382e2046b); Time taken
+--------------+----------------------+
|   category   | average_order_value  |
+--------------+----------------------+
| Electronics  | 26900.0              |
| Furniture    | 12250.0              |
| Stationery   | 1125.0               |
| Unknown      | 15000.0              |
+--------------+----------------------+
4 rows selected (1.009 seconds)
```

Conclusion

This assignment demonstrates a complete data-analysis workflow using Apache Hadoop, HDFS, Apache Hive, and Docker. Transaction data is stored in HDFS and analyzed through an external Hive table using HiveQL. Missing values are handled before calculating total sales revenue and average order value for the product categories.

Docker keeps the required services in a consistent containerized environment, making the overall workflow easier to configure and manage.